

Abdul Rafe Khan

+91-9869620708 | Khanabdulrafe48@gmail.com | github.com/Rafe2001 | [linkedin/Rafe Khan](https://linkedin.com/in/RafeKhan)

EDUCATION

University of Mumbai

B. Tech in Computer Science and Engineering (Data Science)

CGPA: 8.19/10

Mumbai

Aug 2021 – July 2025

EXPERIENCE

AI Engineer, CodeStratLabs

Agentic-AI Chatbot & Computer-Vision

Sep 2025 – Present

- Built and deployed a production-grade **agentic AI airline chatbot** handling end-to-end **flight-related workflows** (booking, meal selection, flight status, baggage, terminal queries) using **LangGraph-based multi-turn orchestration**, with robust **intent management**, **guardrails**, and **system prompts** for Phase-1 release.
- Served as **primary AI developer** for an **Agentic-AI chatbot (ThatCorner)** using **LangGraph**, owning prompts, tools, and orchestration; built **translation pipelines**, **guardrails**, and **multi-turn workflows**, and implemented **memory compression** reducing context from **2,000 to 500 tokens** along with additional **token optimization**, achieving **15–20% LLM cost reduction**.
- Built a **custom LangGraph reducer** to manage post-compression message state; engineered **Redis-backed incremental ingestion pipeline** syncing menu updates across **VectorDB**, **MongoDB**, and Redis cache, and resolved **Amazon Bedrock** strict message sequencing constraints with orphaned `tool_call_id` handling for both Bedrock and OpenAI.
- Developed an end-to-end **computer vision pipeline** for an Israel-based healthcare client automating diagnostic strip classification on a smartphone web app; extracted **9,800 frames**, annotated **2,300 images** on Roboflow, and trained a **YOLOv26-Nano** segmentation model achieving **97.1% Mask mAP@50** (Strip: 99.4%, Pen: 96.6%) at **6.5MB** and **32ms** GPU inference.
- Preprocessed **1,748 strip crops** via perspective transforms and trained a **MobileNetV3-Small** classifier with **99.7% validation accuracy** (Invalid: 99.3%, Negative & Positive: 100%); exported both models PyTorch → ONNX → TF.js (**4× faster** in-browser) and deployed the full detection → crop → classification pipeline in a **React** web app validated on a smartphone browser.

AI Engineer Intern, Techolution [\[Offer Letter\]](#)

RAG, AI Agents, APIs & Prompt Engineering

Feb 2025 – Aug 2025

- Built an AI-driven web-scraping automation and RBAC keyword extractor for Appsecauth **POCs**, later deployed to production.
- Developed and integrated a **WTA/WNTA suggestion API** and **prompt-template retrieval API** within the Instant Learning pipeline, leveraging an intelligent WTA agent for automated validation.
- Contributed to internal **LLM-Studio Libraries** (tools, agents, integrations) to enhance modularity and reuse across projects.
- Designed and optimized **system prompts** for multiple AI agents now deployed in production.

AI Research Intern, IIT Patna [\[Offer Letter\]](#)

Sentiment-Aware ADR Classification

Nov 2023 – Apr 2024

- Scraped the website `askapatient.com` to create around **135,000 rows Hinglish ADR dataset**, followed by annotation and fine-tuning of MuRIL, XLM-RoBERTa, and mBERT models.
- Evaluated model performance using **Precision**, **Recall**, and **F1-score** metrics.
- Automated data scraping for multilingual comments and integrated preprocessing pipelines for code-mixed text.

- Developed an **inventory forecasting model** and **PowerBI dashboard** for Australian and German markets, reducing stockouts through data-driven insights.
- Built a **Q&A chatbot** using Mistral-7B, LangChain, and Llama-Index with ChromaDB; fine-tuned and deployed via Flask API.
- Automated the **USA market costing workflow** and collaborated with Finance and Supply Chain teams for pricing accuracy.
- Created a live **user analytics dashboard** connected to MongoDB for real-time business insights.

SKILLS

- **Programming Languages:** Python, C, Java, SQL
- **AI & ML:** TensorFlow, Keras, PyTorch, Scikit-Learn, LLM's, Llama-index LangChain, LangGraph, OpenCV, Agno, CrewAI
- **Data Science:** : Numpy, Pandas, Matplotlib, Seaborn, PlayWright, Selenium, PowerBI, Ms-Excel, Statistics
- **Web Technologies:** Flask, FastAPI, HTML, CSS, Bootstrap, Postman, Streamlit
- **Databases:** MongoDB, MySQL, PostgreSQL, ChromaDB, AstraDB & other VectorDB's
- **DevOps & Version Control:** Git & GitHub, Docker, Google Cloud (GCP), Jira

PROJECTS

- **CarePlus AI – Multi-Agent Clinic Assistant** GitHub
Streamlit, LangGraph, Groq, PostgreSQL
Developed a multi-agent clinical assistant handling appointment booking, rescheduling, and emergency workflows using LangGraph. Integrated PostgreSQL for persistent data storage and deployed it on Render, enabling real-time chat interactions with email confirmations and calendar scheduling.
- **SuKoon-AI – Mental Health AI Assistant** GitHub
FastAPI, LangGraph, Streamlit, Gemini, Groq, Twilio, Tavily
Designed a production-ready AI mental health assistant integrating LLMs with LangGraph for tool orchestration and RAG for fact-grounded responses. Engineered a strict system prompt to reduce hallucinations and ensure safe, empathetic interactions, with real-time voice support via Twilio.
- **Iris Live – Personal AI Assistant** GitHub
Gradio, OpenCV, SpeechRecognition, ElevenLabs, LangGraph, Python
Built an agentic AI assistant using LangGraph's ReAct architecture with dynamic tool invocation (e.g., webcam) based on voice and visual inputs. Designed system prompts to control tool usage and improve response reliability.
- **Brain Tumor Detection App** GitHub
Flask, OpenCV, TensorFlow, Pillow, CNN
Built a Flask-based web application for brain tumor detection from MRI scans using a custom CNN model, achieving 91% validation accuracy.

ACHIEVEMENTS & EXTRACURRICULAR ACTIVITIES

- **TIAA Retirement Industry AI Hackathon:** Advanced to the *Grand Finale* among 2,200+ participants.
- **Hacknieche Hackathon:** Secured a *Top 10* position in a 36-hour hackathon with 400+ participants.
- **Member, DJS NSS:** Part of the *Creatives Team*, designed posts and stories for the committee's official Instagram handle.